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Using Security Context Constraints (SCC) in OpenShift



Introduction

Security Context Constraints (SCC) in OpenShift are a mechanism to control permissions for pods. They define a set of conditions determining what a pod can and cannot do, particularly related to:

- Running privileged containers
- Accessing the host network or file system
- Running as a specific user or group
- Using specific volume types

SCCs help administrators enforce security policies at the cluster level, ensuring workloads run with the least privilege needed.

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What is an SCC?

An SCC is a cluster-level resource that restricts the security context of pods. When a pod is created, OpenShift checks the SCCs available to the user and assigns the most restrictive SCC that still allows the workload to run.

Key SCC Capabilities

- **RunAsUser:** Defines UID ranges or specific UIDs the container can run as.
 - **SELinuxContext:** Defines SELinux options for pod processes.
 - **FSGroup:** Specifies file system group IDs for volumes.
 - **SupplementalGroups:** Additional groups for processes.
 - **AllowPrivilegedContainer:** Whether privileged mode is allowed.
 - **Volumes:** Which storage volume types are permitted.
 - **Capabilities:** Linux capabilities that can be added or dropped.
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Why SCC is Important in OpenShift

Security in containerized environments requires fine-grained controls to prevent attacks such as privilege escalation, container breakout, or unauthorized host access. SCC helps by:

1. **Minimizing Attack Surface:** Limits pod privileges to only what's necessary.
 2. **Compliance Enforcement:** Aligns workloads with security frameworks (e.g., PCI-DSS, HIPAA).
 3. **Preventing Privilege Escalation:** Blocks containers from running as root unless explicitly allowed.
 4. **Enforcing Consistency:** Ensures all workloads meet security requirements.
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Default SCCs in OpenShift

OpenShift ships with a set of built-in SCCs. Common ones include:

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SCC Name	Purpose
restricted	Most restrictive SCC; blocks privileged mode and enforces UID constraints.
anyuid	Allows running containers as any UID, including root.
privileged	Allows full host access; use with caution.
hostaccess	Allows host networking and ports.
hostnetwork	Allows use of the host's network namespace.
nonroot	Ensures containers run as a non-root UID.
hostmount-anyuid	Allows host mounts without enforcing UID constraints.

Best Practice: Use the **restricted** SCC for most workloads and assign higher-permission SCCs only when absolutely necessary.

Viewing and Managing SCCs

Viewing All SCCs

```
oc get scc
```

Viewing SCC Details

```
oc describe scc restricted
```

This displays rules for the SCC, such as allowed users, SELinux policies, and capabilities.

Assigning SCCs to Users or Service Accounts

Grant SCC to a User

```
oc adm policy add-scc-to-user anyuid developer
```

Grant SCC to a Service Account

```
oc adm policy add-scc-to-user privileged -z default -n mynamespace
```

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Here:

- -z specifies a service account.
- -n specifies the namespace.

Remove SCC from a User

```
oc adm policy remove-scc-from-user anyuid developer
```

How SCCs are Applied

When a pod is created:

1. OpenShift checks all SCCs the user/service account can use.
2. It selects the **most restrictive SCC** that allows the pod's configuration.
3. The pod runs with the restrictions defined by that SCC.

If no SCC allows the requested configuration, the pod creation fails.

SCC vs. Pod Security Standards (PSS)

While Kubernetes offers **Pod Security Standards (PSS)**, OpenShift continues to use SCCs as its primary mechanism for pod-level security.

Feature	SCC	PSS
Enforcement Level	Cluster-specific to OpenShift	Kubernetes-wide standard
Flexibility	Highly customizable	Fixed standards (privileged, baseline, restricted)
Adoption	OpenShift-only	Native Kubernetes

In OpenShift 4.x, SCC remains a critical security component, often coexisting with PSS for hybrid environments.

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Example: Creating a Custom SCC

If you need a custom security policy:

```
apiVersion: security.openshift.io/v1
kind: SecurityContextConstraints
metadata:
  name: custom-scc
allowPrivilegedContainer: false
runAsUser:
  type: MustRunAsRange
  uidRangeMin: 1000
  uidRangeMax: 2000
seLinuxContext:
  type: MustRunAs
  seLinuxOptions:
    user: system_u
    role: system_r
    type: spc_t
    level: s0
fsGroup:
  type: MustRunAs
  ranges:
    - min: 1
      max: 65535
volumes:
- configMap
- secret
users:
- system:serviceaccount:myproject:default
```

Apply the SCC:

```
oc apply -f custom-scc.yaml
```

Best Practices for SCC Usage

- **Principle of Least Privilege:** Use the most restrictive SCC possible.
- **Namespace Separation:** Apply different SCCs per namespace if workloads have varying security needs.
- **Audit Regularly:** Check which users and service accounts have elevated SCCs.
- **Avoid Privileged SCC:** Unless absolutely necessary, avoid giving workloads unrestricted access.
- **Combine with Network Policies:** SCC controls container permissions; network policies control connectivity.

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Troubleshooting SCC Issues

Common SCC-related problems:

- **Pod fails to start with “permission denied” errors:** SCC might be too restrictive for the container's requirements.
- **Unexpected UID in container:** SCC enforces a UID range; container images expecting root may fail.
- **Volume mount denied:** SCC may not allow the volume type (e.g., hostPath).

Resolution Steps:

1. Check pod events using:

```
oc describe pod <pod-name>
```

2. Identify SCC assigned to the pod.
 3. Adjust SCC or container settings accordingly.
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Conclusion

Security Context Constraints are a powerful OpenShift feature for controlling container privileges. By carefully managing SCCs, administrators can ensure workloads operate securely without unnecessary permissions. The key is balancing functionality with security by granting the minimum privileges required.

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